# Coupling the Point Model to the Growth Field

Key points of the meeting of 20 October 2026

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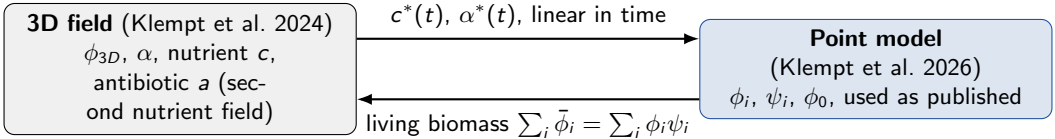
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# What changes

- Point model (Klempt et al. 2026) used as published
- No scaling to the field ($\phi_0 = 1 - \phi_{3D}$ dropped)
- Field $\rightarrow$ point model: $c^*$, $\alpha^*$ from 3D fields
- Point model $\rightarrow$ field: only living biomass consumes
- Eq. 34 with front term, Eq. 36 unchanged

# The loop at one Gauss point



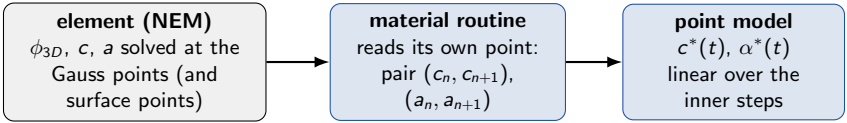

$\sum_i \bar{\phi}_i$ scales the consumption of $c$ and $a$ $\rightarrow$ $c$ enters the front term of Eq. 34 $\rightarrow$ spreading of $\phi_{3D}$ $\rightarrow$ growth $\alpha$ (Eq. 36) $\rightarrow$ stress

# Equations

$$
\begin{aligned}
c^*(t) &= c_0^*\left[\tfrac{t_{n+1}-t}{\Delta t}c_n + \tfrac{t-t_n}{\Delta t}c_{n+1}\right], \quad \alpha^* \text{ likewise} \\
0 &= d\,\nabla^2 c - g\,\phi_{3D}\,\textstyle\sum_i \bar{\phi}_i\, c, \qquad \bar{\phi}_i = \phi_i\psi_i \\
0 &= d_a\nabla^2 a - g_a\,\phi_{3D}\,\textstyle\sum_i \bar{\phi}_i\, a \\
\dot{\phi}_{3D} &= \beta\nabla^2\phi_{3D} + k_\alpha\alpha - \|\nabla\phi_{3D}\|\,\tfrac{r\,c}{k+c}\,\mathbf{n}_{\nabla\phi_{3D}}\cdot\mathbf{n}_{\nabla c}, \qquad \dot{\alpha} = k_\alpha\phi_{3D}
\end{aligned}
$$

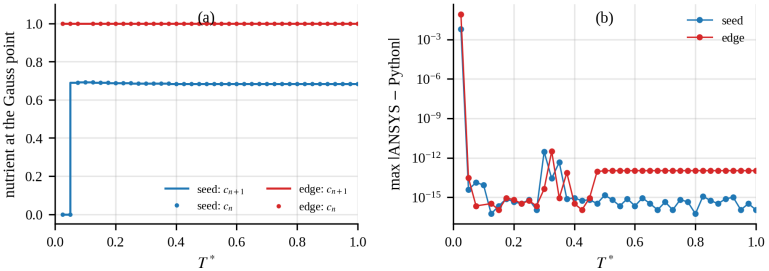
$g\phi$ in Eq. 35 of the paper is a misprint; $g\phi_{3D}c$ is used. $\phi_{3D}$: field, $\phi_i$: point model.

# Nutrient and antibiotic at the Gauss points




- No interpolation from nodes: the fields live at the integration points
- The antibiotic in the second nutrient field takes the same path
- One increment behind: the fields are solved after the material routine
- A short first load step solves the fields before the point model starts

# First ANSYS run: what is checked




- (a) $c_{n+1}$ of one increment is $c_n$ of the next: the pair is passed on without a jump
- (b) point model in ANSYS = Python with the same inputs: $2 \times 10^{-12}$ or better in every step (four runs, with and without antibiotic)
- The first step needs up to 38 Newton iterations from the start state; with a tolerance-controlled Newton (up to 50) both codes agree there too

# Time: one clock for both models

- $T^* = 1$ = duration of the point model's data
- $s$ follows from it, not fitted (synthetic: $s = 0.1$)
- Check (synthetic): error $7 \times 10^{-7}$; with scaling 0.019

# Examples and checks

**Examples** (synthetic, two species)

- A centre, B edges
- A and B mixed
- with / without antibiotic

**Checks**

- equals point model alone
- ANSYS = Python
- $0 \leq \phi \leq 1$
- $\phi_{\text{min}}$, step, mesh

Results to follow

# Assumptions and open points

- Assumptions: linear interpolation, $\sum_i \bar{\phi}_i$ one step late, start values of new points, $\phi_{\text{min}}$
- No transport of species between points
- Front term needs upwinding
- Antibiotic: synthetic data only ($b = 0$ in the calibration)

# Appendix: notation

| | |
|---|---|
| $\phi_{3D}$, $\alpha$ | biofilm phase field ($\phi$ of Klempt et al. 2024) and growth ($\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$), Klempt et al. 2024 |
| $c$, $a$ | nutrient and antibiotic fields (the element's first and second nutrient) |
| $d$, $g$, $d_a$, $g_a$ | diffusion and consumption of nutrient and antibiotic |
| $\beta$, $k_\alpha$, $r$, $k$ | diffusion of $\phi$, growth rate, front growth and Monod constant (Eq. 34, 36) |
| $\phi_i$, $\psi_i$, $\phi_0$ | volume fraction and living fraction of species $i$, auxiliary void (Klempt et al. 2026) |
| $c^*$, $\alpha^*$, $b_i$ | nutrient and antibiotic of the point model, antibiotic sensitivity |
| $\bar{\phi}_i$ | living volume fraction of species $i$, $\phi_i\psi_i$ (Klempt et al. 2026) |
| $T^*$, $s$ | normalised time; point-model time per unit $T^*$ |